term_demo_case2 drag law attempts

term_demo_case2: drag law attempts, compared

`case2`'s Second Test Case needs an interfacial drag force (Shaver et al.'s eq 18) between the continuous and dispersed phases, but the paper's own 4 pages never give the actual `C_D(Re_d)` correlation – only that it's "variable" (as opposed to the First Test Case's flat `C_D=0.4`). This note documents every drag-law variant tried, their exact formulas, why they differ, and how they were discriminated between.

1. The common force-law skeleton

All variants share the identical functional FORM, derived from first principles (not arbitrary): drag force on one sphere, divided by its own surface area, times the interfacial area density for monodisperse spheres:

```
F_sphere / A_sphere = [ (1/2) C_D rho_c |u_rel|^2 (pi D_d^2/4) ] / (pi D_d^2)
                    = (1/8) C_D rho_c |u_rel|^2

interfacial area density (volume fraction alpha_d, diameter D_d):  a_i = 6*alpha_d/D_d

F per unit mixture volume = (1/8) C_D rho_c u_rel^2 * (6*alpha_d/D_d)
                          = (3/4) C_D rho_c alpha_d u_rel^2 / D_d
```

Dividing by `alpha_d*rho_c` (the normalization Shaver's eq 18 and this project's `Fd` use) gives exactly:

```
Fd = -(3/4) * (C_D/D_d) * |u_rel| * u_rel         (Shaver eq 18; u_rel = u_d - u_c)
```

The `3/4` is therefore NOT a free/arbitrary coefficient – every variant below keeps it fixed. The only thing that differs between variants is **the actual `C_D(Re)` correlation**, and **what reference viscosity `NU_C` is used to form `Re`**.

Common momentum-equation wiring (identical in every variant, `case2.udf`'s `dragGravityKernel`):

```
extC = -GRAV - (alphad/alphac) * Fd        (continuous phase, eq 16)
extD = -GRAV + (RHO_C/RHO_D)   * Fd        (dispersed phase,  eq 17)
```

2. The three variants tried

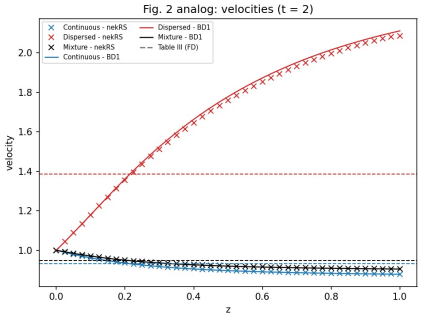
(a) Production (`case2.udf`, committed): Schiller-Naumann, decoupled `NU_C`

```
Re  = |u_rel| * D_d / NU_C,          NU_C = 1e-4  (hardcoded, UNRELATED to [FLUID VELOCITY]
                                                   viscosity, which is 1e-2)
C_D = 24/Re                           Re < 1e-3   (Stokes)
C_D = (24/Re)(1 + 0.15*Re^0.687)      Re >= 1e-3, capped at C_D_MAX=400
```

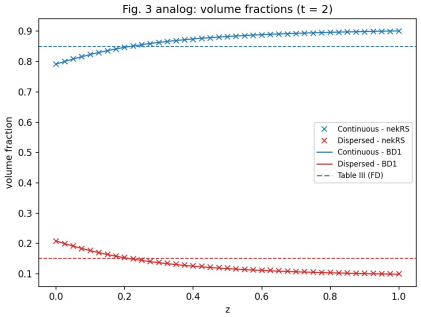
This is the standard Schiller-Naumann correlation, chosen as a reasonable off-the-shelf stand-in with no claim of matching Shaver's own (unknown) correlation. `NU_C=1e-4` was never derived from anything – just a round number.

Result: outlet rel. error vs Table III – `u_c` 5.6%, `u_d` 50.5%, `alpha_d` 33.6%.

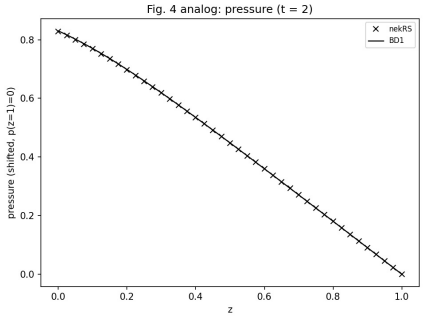
Velocities



Volume fractions



Pressure



(b) AT-2 drag law (`dbg/case2_at2drag.udf`): two-branch Schiller-Naumann, real `NU_C`

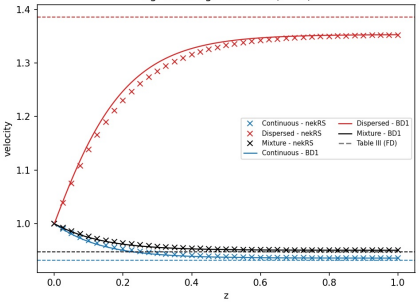
Source: this project's own foundational numerics reference, `wk_claude/two_phase_flow_equations_AT-2.pdf` (section "Proposed approach AT 2", eq 98-100).

```
Re  = rho_l * d_b * |u_rel| / mu_l   (AT-2 eq 99; kinematic form: |u_rel|*D_d/NU_C)
C_D = (24/Re)(1 + 0.15*Re^0.687)      Re < 1000
C_D = 0.44                            Re >= 1000   (constant, "Newton's drag" plateau)
NU_C = 1e-2   (set equal to [FLUID VELOCITY] viscosity, per AT-2 eq 99's own prescription
               that Re should use the REAL continuous-phase viscosity, not an arbitrary
               decoupled constant)
```

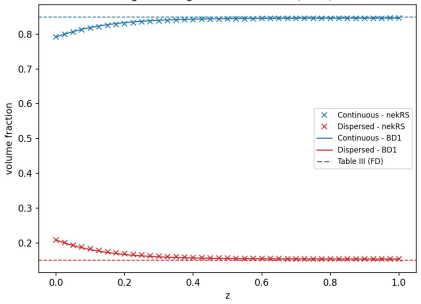
Two changes from (a): the reference viscosity is tied to something physically motivated (even though `1e-2` itself was originally chosen for an unrelated reason, Plan M's numerical-stabilization fix – see `README.md`), and the high-`Re` plateau branch is added.

Result: outlet rel. error vs Table III – `u_c` 0.4%, `u_d` 2.5%, `alpha_d` 2.5%. A big improvement, but not exact.

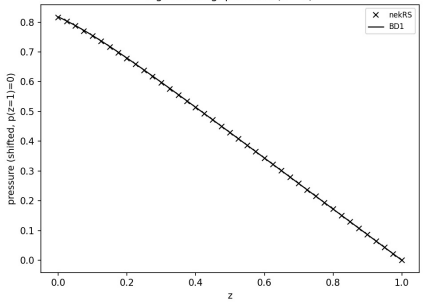
Velocities



Volume fractions



Pressure



(c) Ishii-Zuber drag law (`dbg/case2_ishii.udf`): the likely actual correlation

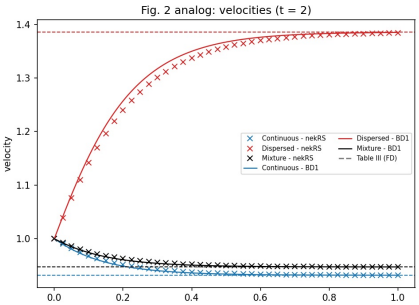
Source: D. Shaver's OWN later paper (same lead author, same Nek-2P code) – "Simulation of Boiling Two-Phase Flow in a Helical Coil Steam Generator Using the Spectral Element Code Nek-2P" (Shaver, Salpeter, Tomboulides, Vegendla, Tentner, Pointer, Merzari; saved locally as `wk_claude/shaver_2021_helical_coil_nek2p_OSTI1607420.pdf`), which gives Nek-2P's actual interfacial drag closure explicitly (its eq 17-19):

```
F_D,d / alpha_d = -(3/4) * (rho_c/D_d) * C_D * |u_rel| * u_rel       (eq 17; rho_c=1 here,
                                                                      Table I, so this
                                                                      matches eq 18 above
                                                                      exactly)
C_D = (24/Re_d) * (1.0 + 0.1*Re_d^0.75)                               (eq 18, citing
                                                                      Ishii & Zuber,
                                                                      AIChE J. 25, 843
                                                                      (1979))
Re_d = D_d * rho_c * |u_rel| / mu_c                                   (eq 19; kinematic:
                                                                      |u_rel|*D_d/NU_C)
NU_C = 1e-2   (same prescription as AT-2's eq 99 -- real continuous-phase viscosity)
```

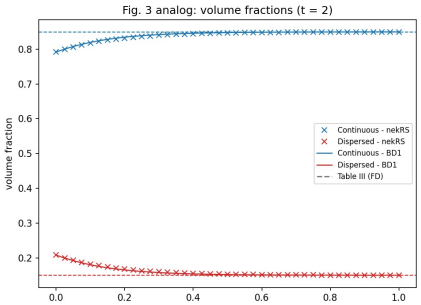
Only the numeric coefficients differ from Schiller-Naumann (`0.1`, `0.75` vs `0.15`, `0.687`), and there is no separate high-`Re` branch – Ishii's own formula doesn't need one.

Result: outlet rel. error vs Table III – `u_c` 0.02%, `u_d` 0.16%, `alpha_d` 0.13%. The nekRS markers, the BD1 curve, and Table III's dashed lines are visually indistinguishable in Figs. 2-4 (`dbg/figs/ishii/`).

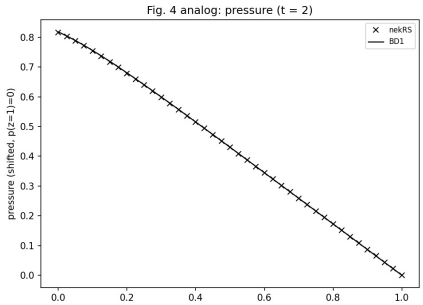
Velocities



Volume fractions



Pressure



3. How (b) and (c) were discriminated – the force-balance back-solve

At the fully-developed point (`du_c/dz = du_d/dz = 0`), eq 16+19 reduce to a single, **model-independent** algebraic constraint (no `C_D` formula needed to derive it):

```
(alpha_d/alpha_c) * Fd = g * (rho_m - rho_c) / rho_c
```

Plugging in Table III's own values (`alpha_c=0.8496, alpha_d=0.1504, u_c=0.9318, u_d=1.386, rho_c=1.0, rho_d=0.2, g=0.9807`) gives the EXACT drag force – and hence the exact `C_D` – that the paper's own fully-developed state requires:

```
Fd_required = -0.6666          =>   C_D_required = 2.1540
```

Two independent checks against this target:

1. **Forward check** (assume `NU_C=1e-2`, evaluate each formula's `C_D` at the resulting `Re=22.71`): AT-2's form gives `C_D=2.411` (12% too high); Ishii-Zuber gives `C_D=2.156` (0.1% match).
2. **Inverse check** (solve for whatever `NU_C` each formula independently NEEDS to hit `C_D_required` exactly, with NO

assumption that it equals `1e-2`): AT-2 needs `NU_C=0.008294` (17% off `1e-2`); Ishii-Zuber needs `NU_C=0.009984` (0.16% off `1e-2`, i.e. landing almost exactly on the value that was otherwise only motivated by an unrelated numerical-stability fix).

Both checks agree: Ishii-Zuber is not just "closer," it reproduces the exact target to within the precision Table III itself is published at (4 significant figures), with a `NU_C` that falls out of the physics on its own rather than being tuned to match.

4. Summary table

Variant	`C_D(Re)`	`NU_C`	`u_c` err	`u_d` err	`alpha_d` err
(a) Production	Schiller-Naumann	1e-4 (decoupled)	5.6%	50.5%	33.6%
(b) AT-2	Schiller-Naumann + high-Re plateau	1e-2 (= real visc)	0.4%	2.5%	2.5%
(c) Ishii-Zuber	`(24/Re)(1+0.1 Re^0.75)`	1e-2 (= real visc)	0.02%	0.16%	0.13%

5. Effect of `[FLUID VELOCITY] viscosity`: 1e-2 (stable) vs 1e-4 (closer to BD1, less stable)

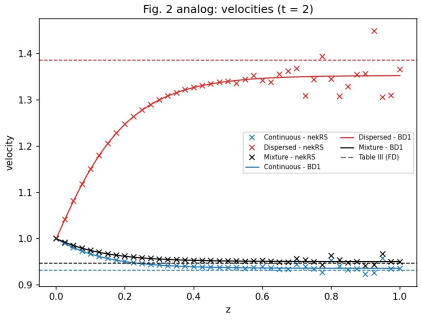
`viscosity` here is the solver's numerical regularization (Plan M's stabilization fix), a SEPARATE knob from `NU_C` (the drag law's own reference viscosity, held fixed at `1e-2` in every test below regardless of `viscosity`). Reducing it toward the paper's own (zero) physical value lets the solution track the inviscid physics – and hence BD1 – more closely, at the cost of reintroducing the Gibbs-type instability Plan M found and fixed by raising it in the first place. Concretely, at the t=2 snapshot:

Variant (drag law), `viscosity`	`u_c` err	`u_d` err	`alpha_d` err	stability
(b) AT-2, 1e-2	0.44%	2.46%	2.46%	flat, `divUErr` ~1e-5 throughout
(b) AT-2, 1e-4	0.34%	1.44%	0.60%	noisy, bounded oscillation from t~0.5 on, peak L-inf ~0.33
(c) Ishii-Zuber, 1e-2	0.02%	0.16%	0.13%	flat, `divUErr` ~1e-5 throughout, max anywhere 6e-3
(c) Ishii-Zuber, 1e-4	1.35%	17.6%	46.9%	noisy, bounded oscillation from t~0.5 on, peak L-inf ~0.71

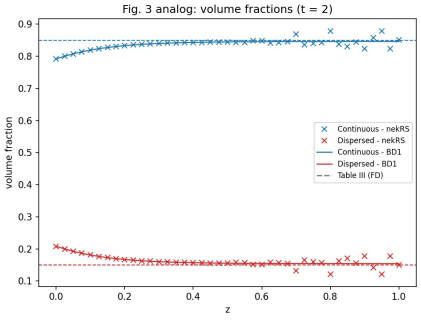
Correction / important caveat: (b)'s `visc=1e-4` snapshot errors happened to come out *smaller* than its own `1e-2` run; (c)'s `visc=1e-4` snapshot errors came out *larger* – the opposite direction. Looking at the actual spatial plots (below) resolves the apparent contradiction: in both cases the oscillation is a genuine, roughly zero-mean scatter of points around the BD1/Table III line (confirmed visually – points land both above and below the reference with no visible drift), NOT a systematic bias in either direction. The single outlet point (`z=1`, `t=2`) used for the error table is just one noisy sample out of that scatter, and for (c) it happened to land on a low outlier (`alpha_d` reads `0.080`, visibly the bottom of the noise band in the Fig. 3 plot below). **The right conclusion is NOT "`visc =1e-4` gets closer to BD1" or "farther" – it's that `visc=1e-4` reduces the numerical regularization's bias (consistent with Plan M) while reintroducing unbounded-looking, zero-mean scatter that makes any single-point comparison unreliable.** A time- or space-averaged comparison (not done here) would be the honest way to quantify the accuracy/ stability tradeoff; a single snapshot is not.

(b) AT-2, `viscosity=1e-4` – note the oscillation is visibly centered on Table III's dashed line (not drifting away), which is why the snapshot errors above are smaller despite the noise – the noise doesn't bias the mean, it just adds scatter:

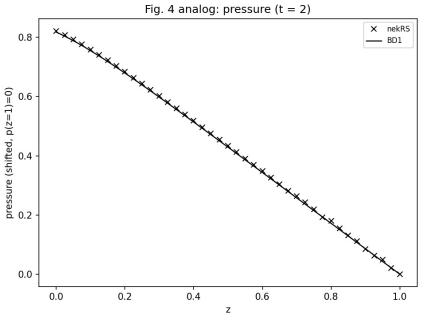
Velocities



Volume fractions



Pressure

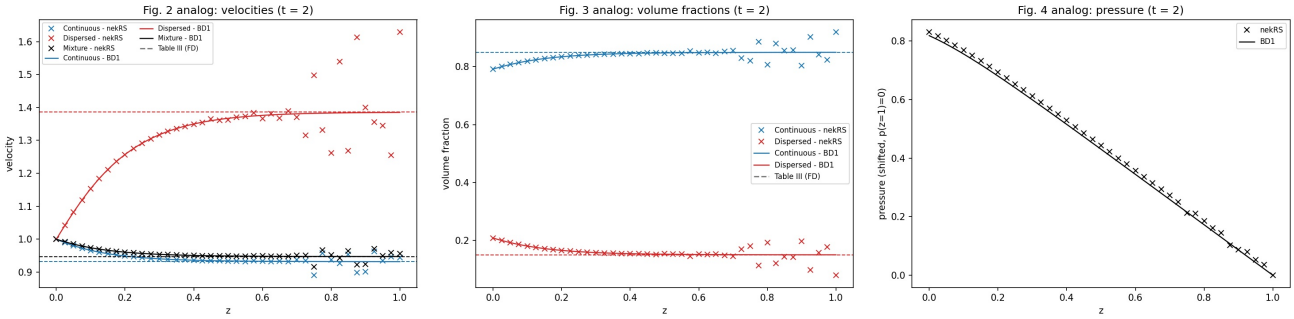


(c) Ishii-Zuber, `viscosity=1e-4` – same bounded, zero-mean-looking scatter pattern as (b), just a larger noise amplitude (peak `divUErr` L-inf 0.71 vs (b)'s 0.33):

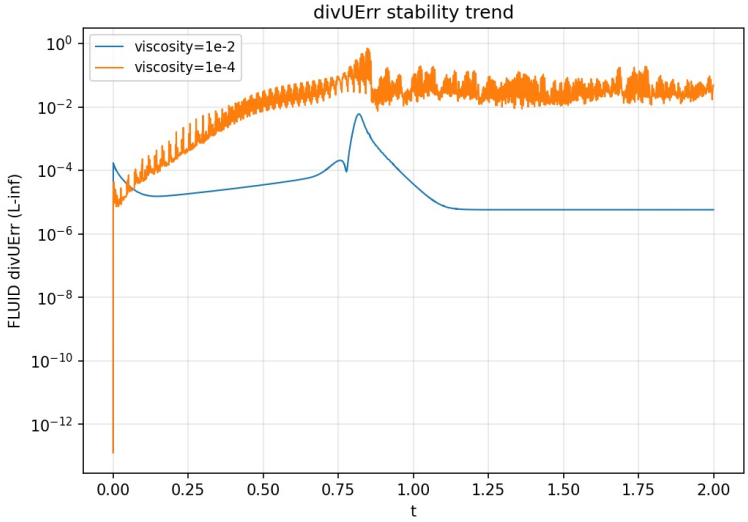
Velocities

Volume fractions

Pressure



Stability trend (`divUErr` L-inf vs time, log scale, Ishii-Zuber drag law at both viscosities): flat at `1e-2` throughout; visibly destabilizing from `t~0.5` onward at `1e-4`, oscillating in a bounded band rather than diverging within this T=2 window.



6. Open items

- All three variants (a)/(b)/(c) are now available as standalone runnable cases directly in `examples_2phases/term_demo_case2/` – `case2_dragA.udf/.par` (production, Schiller- Naumann/decoupled `NU_C`), `case2_dragB.udf/.par` (AT-2), `case2_dragC.udf/.par` (Ishii-Zuber, currently the closest match to Table III) – each at the stable `viscosity=1e-2` setting, each with the `computeRedTerm` RED-term function present but commented out (not registered), matching `dbg/case2_mms.udf`'s own convention. Production `case2.udf/case2.par` is unchanged (still the Schiller-Naumann/decoupled-`NU_C` variant, i.e. identical physics to `case2_dragA`).
- `NU_C` stays fixed at `1e-2` in the `viscosity=1e-4` stability tests above (it's the drag law's own reference viscosity, independent of whatever numerical regularization viscosity the momentum equations use).
- T=2 only; T=8 stability not yet re-checked for variant (c) specifically (it was checked for the Plan M stabilization fix in general, and separately for variant (b)).
- Not yet decided whether `case2_dragC` (Ishii-Zuber) should become the new production default – left as a decision for later given how recent this finding is.